

P118

3. 试用与非门设计一个逻辑电路。当三个输入A、B、C中至少有两个为低时，该电路输出为高。

解：

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

根据真值表可写出函数的最小项表达式为

$$F(A, B, C) = \sum m(0, 1, 2, 4)$$

对应的卡诺图如下：

BC	00	01	11	10
A=0	1	1	0	1
A=1	1	0	0	0

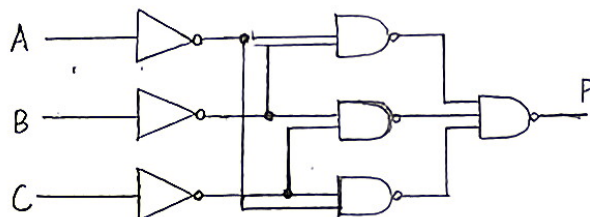
得最简与-或式 $F = \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{A}\bar{B}$

$$\bar{F} = \overline{\bar{A}\bar{C} + \bar{B}\bar{C} + \bar{A}\bar{B}} = \overline{\bar{A}\bar{C}} \cdot \overline{\bar{B}\bar{C}} \cdot \overline{\bar{A}\bar{B}}$$

$$\bar{F} = \overline{\bar{A}\bar{C}} \cdot \overline{\bar{B}\bar{C}} \cdot \overline{\bar{A}\bar{B}}$$

$$\text{两次取反, 得 } F = \overline{\overline{\bar{A}\bar{C}} \cdot \overline{\bar{B}\bar{C}} \cdot \overline{\bar{A}\bar{B}}}$$

逻辑电路图：



4. 在有原变量又有反变量的输入条件下，分别用与非门、或非门设计实现下列函数的组合电路

$$(1) F(A, B, C, D) = \sum m(0, 2, 3, 4, 5, 6, 7, 12, 14, 15)$$

解：对应的卡诺图如下：

AB \ CD	00	01	11	10
00	1	0	1	1
01	1	1	1	1
11	1	0	1	1
10	0	0	0	0

① 用与非门

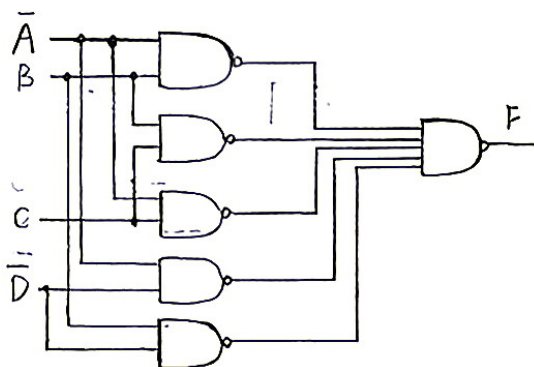
$$\text{得最简与-或式 } F = \bar{A}B + \bar{A}C + BC + \bar{A}\bar{D} + \bar{B}\bar{D}$$

$$\text{经 } \bar{F} = \overline{\bar{A}B + \bar{A}C + BC + \bar{A}\bar{D} + \bar{B}\bar{D}} = \overline{\bar{A}B} \cdot \overline{\bar{A}C} \cdot \overline{BC} \cdot \overline{\bar{A}\bar{D}} \cdot \overline{\bar{B}\bar{D}}$$

$$\bar{F} = \overline{\bar{A}B} \cdot \overline{\bar{A}C} \cdot \overline{BC} \cdot \overline{\bar{A}\bar{D}} \cdot \overline{\bar{B}\bar{D}}$$

$$\text{两次取反, 得 } F = \overline{\overline{\bar{A}B} \cdot \overline{\bar{A}C} \cdot \overline{BC} \cdot \overline{\bar{A}\bar{D}} \cdot \overline{\bar{B}\bar{D}}}$$

逻辑电路图：



取反 $\rightarrow F = \bar{F} = (B+C+\bar{D})(\bar{A}+C+\bar{D})(A+B)$

② 用或非门: $\bar{F} = \bar{B}\bar{C}D + A\bar{C}D + AB$

AB \ CD	00	01	11	10
00	1	0	1	1
01	1	1	1	1
11	1	0	1	1
10	0	0	0	0

用0格化简逻辑函数, 得最简或与表达式,

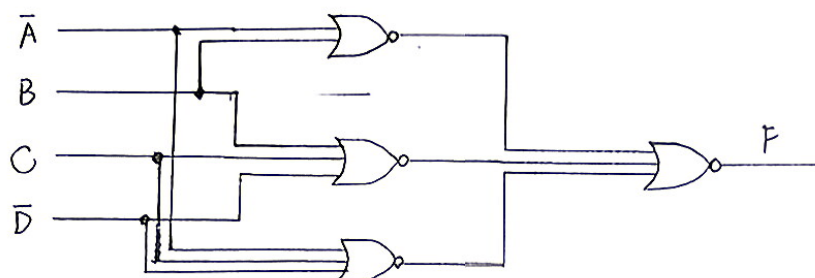
$$F = (\bar{A}+B)(\bar{A}+C+\bar{D})(B+C+\bar{D})$$

经两次取反, 得

$$F = \bar{\bar{F}} = \overline{(\bar{A}+B)(\bar{A}+C+\bar{D})(B+C+\bar{D})}$$

$$= \overline{\bar{A}+B + \bar{A}+C+\bar{D} + B+C+\bar{D}}$$

逻辑电路图如下:



(2) $F(A,B,C,D) = \sum m(0,2,5,9,15) + \sum \phi(6,7,8,10,12,13)$

解: ① 用与非门:

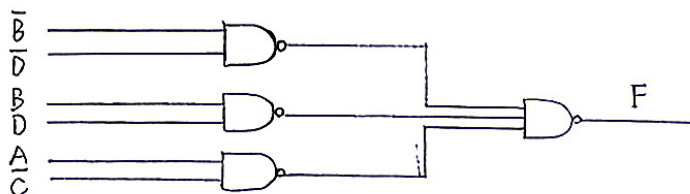
AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	x	x
11	x	x	1	0
10	x	1	0	x

由卡诺图得最简与-或式 $F = \bar{B}\bar{D} + BD + A\bar{C}$

经两次取反, 得

$$F = \bar{\bar{F}} = \overline{\bar{B}\bar{D} + BD + A\bar{C}} = \overline{\bar{B}\bar{D} \cdot BD \cdot A\bar{C}}$$

逻辑电路图如下:



② 用或非门:

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	x	x
11	x	x	1	0
10	x	1	0	x

用0格化简逻辑函数, 得最简或与表达式

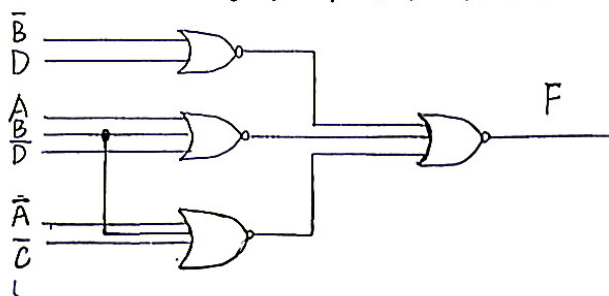
$$F = (\bar{B}+D)(A+B+\bar{D})(\bar{A}+B+\bar{C})$$

经两次取反, 得

$$F = \bar{\bar{F}} = \overline{(\bar{B}+D)(A+B+\bar{D})(\bar{A}+B+\bar{C})}$$

$$= \overline{\bar{B}+D + A+B+\bar{D} + \bar{A}+B+\bar{C}}$$

逻辑电路图如下:



8. 设计一个编码器, 7个输入信号和输出的3位代码之间的对应关系如下所示

X_0	X_1	X_2	X_3	X_4	X_5	X_6	U	V	W
0	1	1	1	1	1	1	0	0	1
1	0	1	1	1	1	1	0	1	0
1	1	0	1	1	1	1	0	1	1
1	1	1	0	1	1	1	1	0	0
1	1	1	1	0	1	1	1	0	1
1	1	1	1	1	0	1	1	1	0
1	1	1	1	1	1	0	1	1	1

若令输入信号有效为1, 无效为0

$X_0 X_1 X_2 \bar{X}_3 X_4 X_5 X_6 \dots$ 又 $\bar{X}_3 = X_0 X_1 X_2 X_4 X_5 X_6 \therefore \bar{X}_3 = \bar{X}_3 \bar{X}_3 = \bar{X}^3$

则 $U = X_0 X_1 X_2 \bar{X}_3 X_4 X_5 X_6 + X_0 X_1 X_2 X_3 \bar{X}_4 X_5 X_6 + X_0 X_1 X_2 X_3 X_4 \bar{X}_5 X_6 + X_0 X_1 X_2 X_3 X_4 X_5 \bar{X}_6$

$V = X_0 \bar{X}_1 X_2 X_3 X_4 X_5 X_6 + X_0 X_1 \bar{X}_2 X_3 X_4 X_5 X_6 + X_0 X_1 X_2 \bar{X}_3 X_4 X_5 X_6 + X_0 X_1 X_2 X_3 \bar{X}_4 X_5 X_6$

$W = \bar{X}_0 X_1 X_2 X_3 X_4 X_5 X_6 + X_0 \bar{X}_1 \bar{X}_2 X_3 X_4 X_5 X_6 + X_0 X_1 \bar{X}_2 X_3 \bar{X}_4 X_5 X_6 + X_0 X_1 X_2 \bar{X}_3 X_4 X_5 \bar{X}_6$

以表达式U为例简化上述逻辑表达式

$U = \bar{X}_3 + \bar{X}_4 + \bar{X}_5 + \bar{X}_6$

若两个逻辑变量满足如下方程 $\begin{cases} x+y=1 \\ x \cdot y=0 \end{cases}$, 则有 $x = \bar{y}$

$X_3 = \overline{X_0 + X_1 + X_2 + X_4 + X_5 + X_6} = \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_4 \bar{X}_5 \bar{X}_6$

$X_4 = \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_5 \bar{X}_6$

$X_5 = \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 \bar{X}_6$

$X_6 = \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 \bar{X}_5$

代入U, 得 $U = \bar{X}_3 + \bar{X}_4 + \bar{X}_5 + \bar{X}_6$, 同理, 得: $V = \bar{X}_1 + \bar{X}_2 + \bar{X}_5 + \bar{X}_6$, $W = \bar{X}_0 + \bar{X}_2 + \bar{X}_4 + \bar{X}_6$

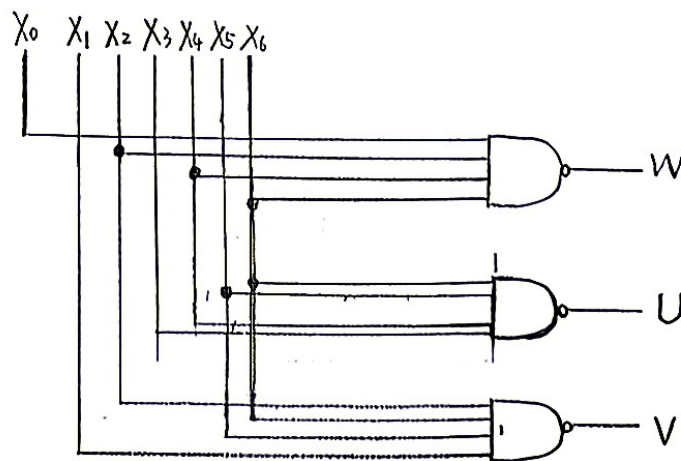
若用与非门实现上述逻辑方程, 则

$U = \bar{U} = \overline{\bar{X}_3 + \bar{X}_4 + \bar{X}_5 + \bar{X}_6} = \overline{\bar{X}_3 \bar{X}_4 \bar{X}_5 \bar{X}_6}$

$V = \bar{V} = \overline{\bar{X}_1 + \bar{X}_2 + \bar{X}_5 + \bar{X}_6} = \overline{\bar{X}_1 \bar{X}_2 \bar{X}_5 \bar{X}_6}$

$W = \bar{W} = \overline{\bar{X}_0 + \bar{X}_2 + \bar{X}_4 + \bar{X}_6} = \overline{\bar{X}_0 \bar{X}_2 \bar{X}_4 \bar{X}_6}$

逻辑电路图如下:



若令输入信号有效为0, 无效为1

$$U = \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 \bar{X}_5 \bar{X}_6 + \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 X_4 \bar{X}_5 \bar{X}_6 + \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 X_5 \bar{X}_6 + \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 \bar{X}_5 X_6$$

$$V = \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 \bar{X}_5 \bar{X}_6 + \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 \bar{X}_5 \bar{X}_6 + \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 X_5 \bar{X}_6 + \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 \bar{X}_5 X_6$$

$$W = X_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 \bar{X}_5 \bar{X}_6 + \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 \bar{X}_5 \bar{X}_6 + \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 X_4 \bar{X}_5 \bar{X}_6 + \bar{X}_0 \bar{X}_1 \bar{X}_2 \bar{X}_3 \bar{X}_4 X_5 \bar{X}_6$$

以表达式U为例简化上述逻辑表达式

若两个逻辑变量满足如下方程 $\begin{cases} x+y=1 \\ x \cdot y=0 \end{cases}$, 则有 $x=\bar{y}$

$$X_3 = \overline{\bar{X}_0 + \bar{X}_1 + \bar{X}_2 + \bar{X}_4 + \bar{X}_5 + \bar{X}_6} = X_0 X_1 X_2 X_4 X_5 X_6$$

$$X_4 = X_0 X_1 X_2 X_3 X_5 X_6$$

$$X_5 = X_0 X_1 X_2 X_3 X_4 X_6$$

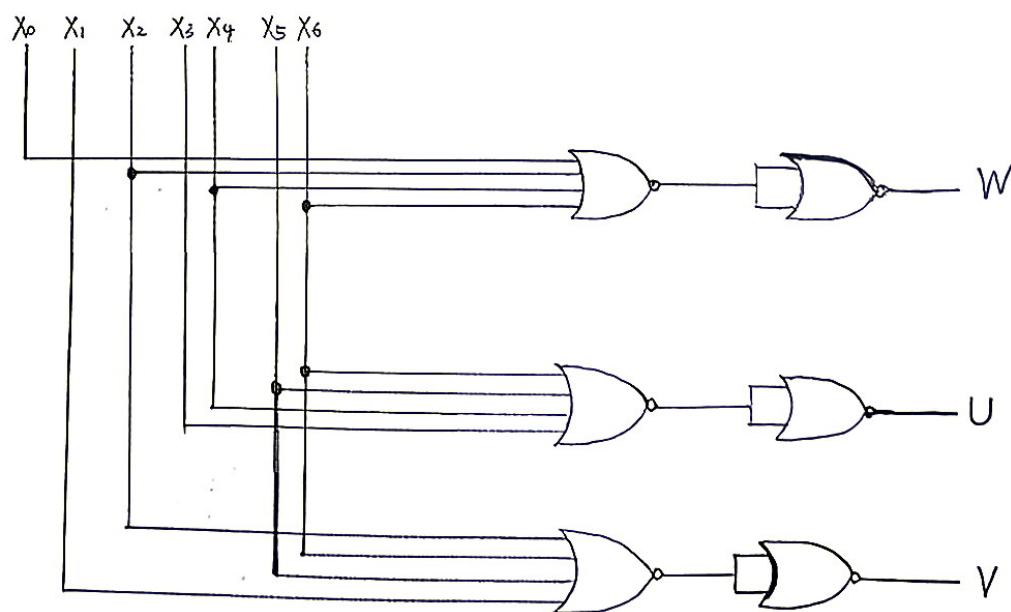
$$X_6 = X_0 X_1 X_2 X_3 X_4 X_5$$

代入U, 得 $U = X_3 + X_4 + X_5 + X_6$

同理可得 $V = X_1 + X_2 + X_5 + X_6$

$$W = X_0 + X_2 + X_4 + X_6$$

若用或非门实现上述逻辑方程, 则可作逻辑电路图如下:



9. 用与非门设计实现 8421 BCD 码转换为 2421 BCD 码的译码电路, 其转换码表如下

$(N)_{10}$	8421 码				2421 码			
	X_0	X_1	X_2	X_3	A_0	A_1	A_2	A_3
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	0
3	0	0	1	1	0	0	1	1
4	0	1	0	0	0	1	0	0
5	0	1	0	1	1	0	1	1
6	0	1	1	0	1	1	0	0
7	0	1	1	1	1	1	0	1
8	1	0	0	0	1	1	1	0
9	1	0	0	1	1	1	1	1

解:

$X_0 X_1$	$X_2 X_3$			
	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	x	x	x	x
10	1	1	x	x

A_0

$X_0 X_1$	$X_2 X_3$			
	00	01	11	10
00	0	0	0	0
01	1	0	1	1
11	x	x	x	x
10	1	1	x	x

A_1

$X_0 X_1$	$X_2 X_3$			
	00	01	11	10
00	0	0	1	1
01	0	1	0	0
11	x	x	x	x
10	1	1	x	x

A_2

$X_0 X_1$	$X_2 X_3$			
	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	x	x	x	x
10	0	1	x	x

A_3

$$A_0 = X_0 + X_1 X_3 + X_1 X_2$$

$$= \overline{\overline{X_0} \cdot \overline{X_1 X_3} \cdot \overline{X_1 X_2}}$$

$$A_1 = X_0 + X_1 X_2 + X_1 \bar{X}_3$$

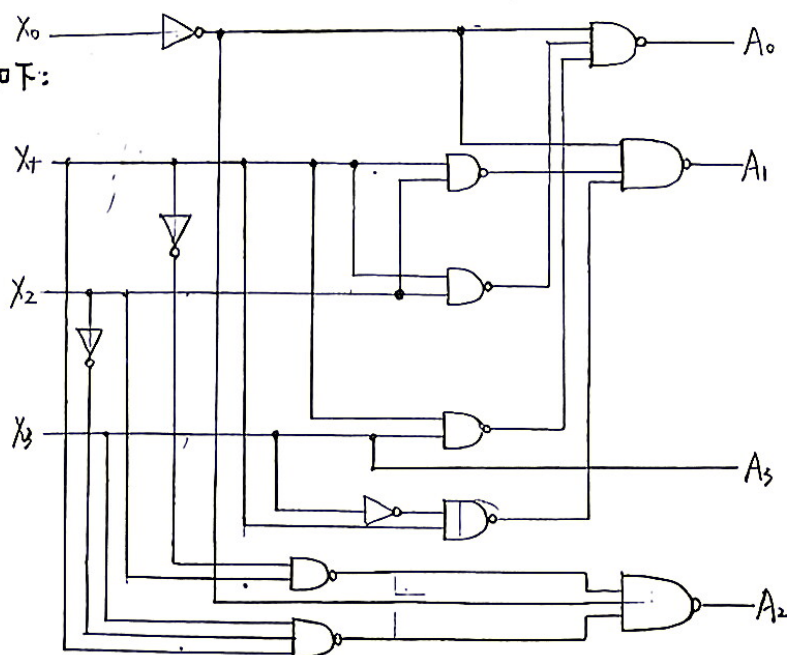
$$= \overline{\overline{X_0} \cdot \overline{X_1 X_2} \cdot \overline{X_1 \bar{X}_3}}$$

$$A_2 = X_0 + \bar{X}_1 X_2 + X_1 \bar{X}_2 X_3$$

$$= \overline{\overline{X_0} \cdot \overline{\bar{X}_1 X_2} \cdot \overline{X_1 \bar{X}_2 X_3}}$$

$$A_3 = X_3$$

逻辑电路图如下:

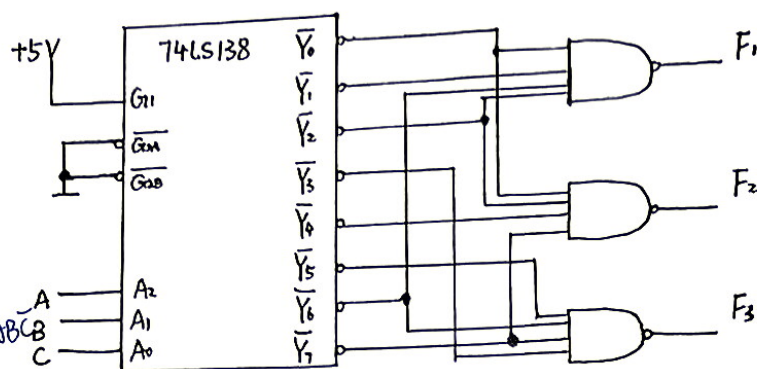


10. 分析所示逻辑电路图, 写出 F_1 、 F_2 、 F_3 的逻辑函数表达式

$$F_1 = \overline{Y_0} \cdot \overline{Y_1} \cdot \overline{Y_2} \cdot \overline{Y_6}$$

$$= Y_0 + Y_1 + Y_2 + Y_6$$

$$= \overline{A_2}\overline{A_1}\overline{A_0} + \overline{A_2}\overline{A_1}A_0 + \overline{A_2}A_1\overline{A_0} + A_2\overline{A_1}\overline{A_0}$$



解: $F_1 = \overline{Y_0} \overline{Y_1} \overline{Y_2} \overline{Y_6} = Y_0 + Y_1 + Y_2 + Y_6 = \overline{A_2}\overline{A_1}\overline{A_0} + \overline{A_2}\overline{A_1}A_0 + \overline{A_2}A_1\overline{A_0} + A_2\overline{A_1}\overline{A_0}$

$$F_2 = \overline{Y_0} \overline{Y_2} \overline{Y_4} \overline{Y_7} = Y_0 + Y_2 + Y_4 + Y_7 = \overline{A_2}\overline{A_1}\overline{A_0} + \overline{A_2}A_1\overline{A_0} + A_2\overline{A_1}\overline{A_0} + A_2A_1\overline{A_0}$$

$$F_3 = \overline{Y_3} \overline{Y_5} \overline{Y_6} \overline{Y_7} = Y_3 + Y_5 + Y_6 + Y_7 = \overline{A_2}A_1\overline{A_0} + A_2\overline{A_1}\overline{A_0} + A_2A_1\overline{A_0} + A_2A_1A_0$$

11. 用3线-8线译码器实现函数 $F = AB + BC + AC$ 的逻辑电路

解: 真值表如下:

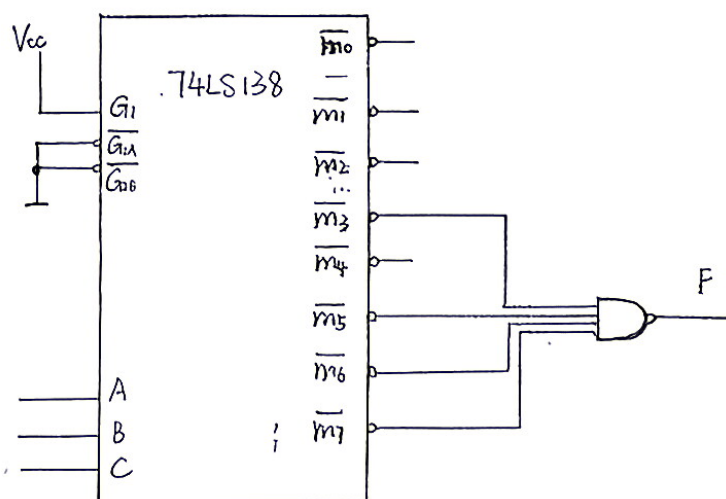
A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$F = \sum m(3, 5, 6, 7)$$

$$= m_3 + m_5 + m_6 + m_7$$

$$= \overline{m_3} \cdot \overline{m_5} \cdot \overline{m_6} \cdot \overline{m_7}$$

逻辑电路图如下:



13. 用两片双四选-数据选择器和与非门实现8421 BCD码至2421 BCD码的转换

解: 设8421 BCD码输入为 $B_4 B_3 B_2 B_1$, 2421 BCD码输出为 $F_4 F_3 F_2 F_1$

则有真值表:

B_4	B_3	B_2	B_1	F_4	F_3	F_2	F_1
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	1
0	1	0	0	0	1	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	1	0	0
0	1	1	1	1	1	0	1
1	0	0	0	1	1	1	0
1	0	0	1	1	1	1	1

$$W = \sum_{i=0}^3 m_i D_i$$

$$F_4 = \bar{B}_4 \bar{B}_3 \bar{B}_2 \bar{B}_1 + \bar{B}_4 \bar{B}_3 \bar{B}_2 B_1 + \bar{B}_4 \bar{B}_3 B_2 \bar{B}_1 + \bar{B}_4 \bar{B}_3 B_2 B_1 + B_4 \bar{B}_3 \bar{B}_2 \bar{B}_1 + B_4 \bar{B}_3 \bar{B}_2 B_1$$

$$F_3 = \bar{B}_4 \bar{B}_3 \bar{B}_2 \bar{B}_1 + \bar{B}_4 \bar{B}_3 \bar{B}_2 B_1 + \bar{B}_4 \bar{B}_3 B_2 \bar{B}_1 + \bar{B}_4 \bar{B}_3 B_2 B_1 + B_4 \bar{B}_3 \bar{B}_2 \bar{B}_1 + B_4 \bar{B}_3 \bar{B}_2 B_1$$

$$F_2 = \bar{B}_4 \bar{B}_3 \bar{B}_2 \bar{B}_1 + \bar{B}_4 \bar{B}_3 \bar{B}_2 B_1 + \bar{B}_4 \bar{B}_3 B_2 \bar{B}_1 + \bar{B}_4 \bar{B}_3 B_2 B_1 + B_4 \bar{B}_3 \bar{B}_2 \bar{B}_1 + B_4 \bar{B}_3 \bar{B}_2 B_1$$

$$F_1 = \bar{B}_4 \bar{B}_3 \bar{B}_2 \bar{B}_1 + \bar{B}_4 \bar{B}_3 \bar{B}_2 B_1 + \bar{B}_4 \bar{B}_3 B_2 \bar{B}_1 + \bar{B}_4 \bar{B}_3 B_2 B_1 + B_4 \bar{B}_3 \bar{B}_2 \bar{B}_1 + B_4 \bar{B}_3 \bar{B}_2 B_1$$

已知4选1 MUX 只有两个选择控制端 $A_1 A_0$, 因此选 B_2, B_1 作为选择控制端, B_3, B_4 放在数据输入端

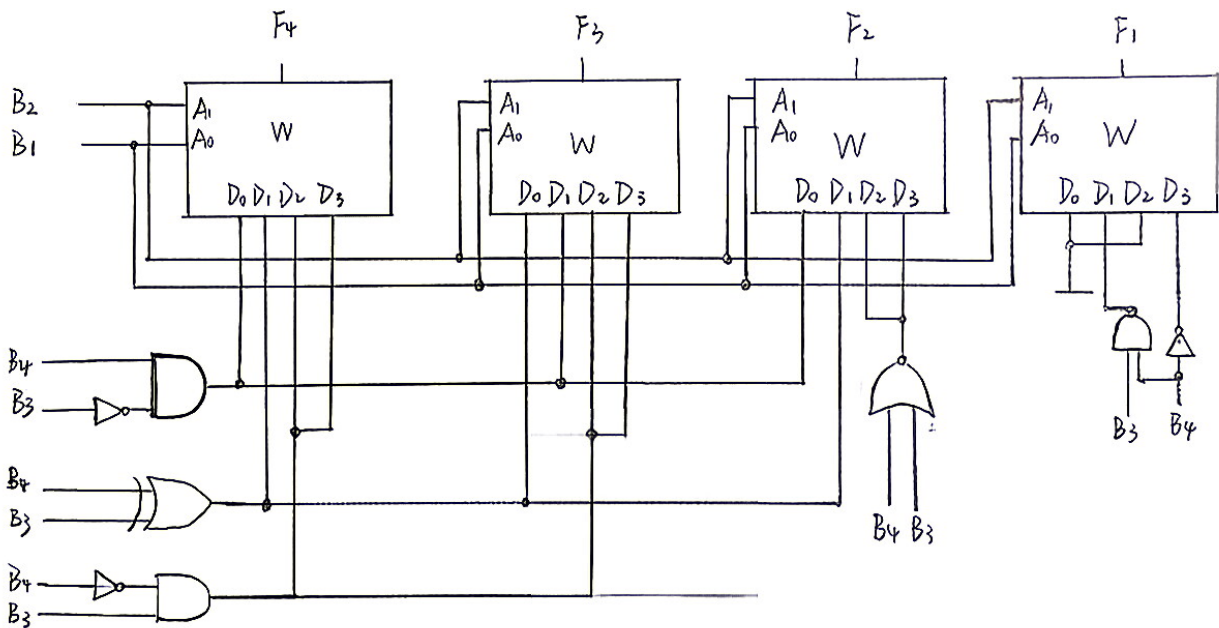
对 F_4, F_3, F_2, F_1 作变换:

$$F_4 = \bar{B}_2 \bar{B}_1 \cdot (B_4 \bar{B}_3) + \bar{B}_2 \bar{B}_1 \cdot (B_4 \oplus B_3) + \bar{B}_2 \bar{B}_1 \cdot (\bar{B}_4 B_3) + \bar{B}_2 \bar{B}_1 \cdot (\bar{B}_4 B_3)$$

$$F_3 = \bar{B}_2 \bar{B}_1 \cdot (B_4 \oplus B_3) + \bar{B}_2 \bar{B}_1 \cdot (B_4 \bar{B}_3) + \bar{B}_2 \bar{B}_1 \cdot (\bar{B}_4 B_3) + \bar{B}_2 \bar{B}_1 \cdot (\bar{B}_4 B_3)$$

$$F_2 = \bar{B}_2 \bar{B}_1 \cdot (B_4 \bar{B}_3) + \bar{B}_2 \bar{B}_1 \cdot (B_4 \oplus B_3) + \bar{B}_2 \bar{B}_1 \cdot (\bar{B}_4 B_3) + \bar{B}_2 \bar{B}_1 \cdot (\bar{B}_4 B_3)$$

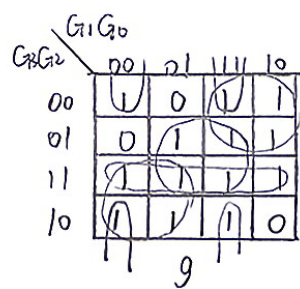
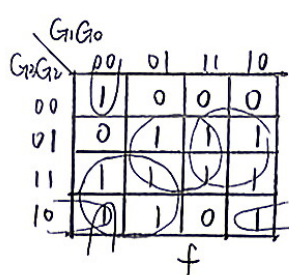
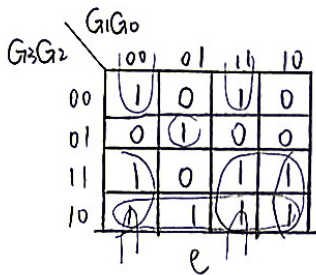
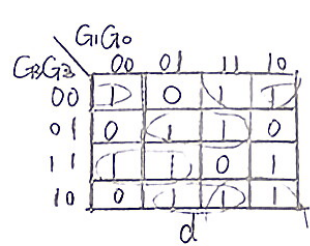
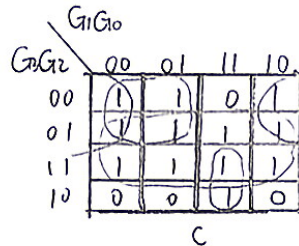
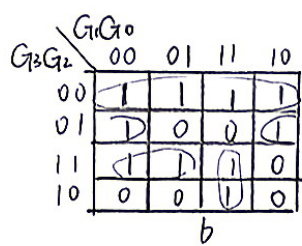
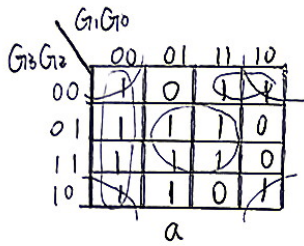
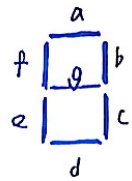
$$F_1 = \bar{B}_2 \bar{B}_1 \cdot (\bar{B}_4 + B_3) + \bar{B}_2 \bar{B}_1 \cdot B_4$$



17. 设计实现将格雷码转换成七段数字(共阴)显示代码的转换译码电路

解:

G_3	G_2	G_1	G_0	a	b	c	d	e	f	g	字形
0	0	0	0	1	1	1	1	1	1	0	0
0	0	0	1	0	1	1	0	0	0	0	1
0	0	1	1	1	1	0	1	1	0	1	2
0	0	1	0	1	1	1	1	0	0	1	3
0	1	1	0	0	1	1	0	0	1	1	4
0	1	1	1	1	0	1	1	0	1	1	5
0	1	0	1	1	0	1	1	1	1	1	6
0	1	0	0	1	1	1	1	0	0	0	7
1	1	0	0	1	1	1	1	1	1	1	8
1	1	0	1	1	1	1	1	0	1	1	9
1	1	1	1	1	1	1	0	1	1	1	A
1	1	1	0	0	0	1	1	1	1	1	B
1	0	1	0	1	0	0	1	1	1	0	C
1	0	1	1	0	1	1	1	1	0	1	D
1	0	0	1	1	0	0	1	1	1	1	E
1	0	0	0	1	0	0	0	1	1	1	F



$$a = G_2 G_0 + \overline{G_3} \overline{G_2} G_1 + \overline{G_2} \overline{G_1} + \overline{G_1} \overline{G_0}$$

$$b = \overline{G_3} \overline{G_2} + \overline{G_3} G_2 \overline{G_0} + G_3 G_2 \overline{G_1} + G_3 G_1 G_0$$

$$c = G_2 + G_3 G_1 G_0 + \overline{G_3} \overline{G_1} + \overline{G_3} \overline{G_0}$$

$$d = \overline{G_2} G_1 + \overline{G_3} \overline{G_2} \overline{G_0} + \overline{G_3} G_2 G_0 + G_3 G_2 \overline{G_1} + G_3 \overline{G_2} G_0$$

$$e = G_3 \overline{G_0} + G_3 G_1 + G_3 \overline{G_2} + \overline{G_3} G_2 \overline{G_1} G_0 + \overline{G_2} \overline{G_1} \overline{G_0} + \overline{G_0} G_1 G_0$$

$$f = G_2 G_1 + G_2 G_0 + G_3 \overline{G_1} + \overline{G_2} \overline{G_1} \overline{G_0} + G_3 \overline{G_2} \overline{G_0}$$

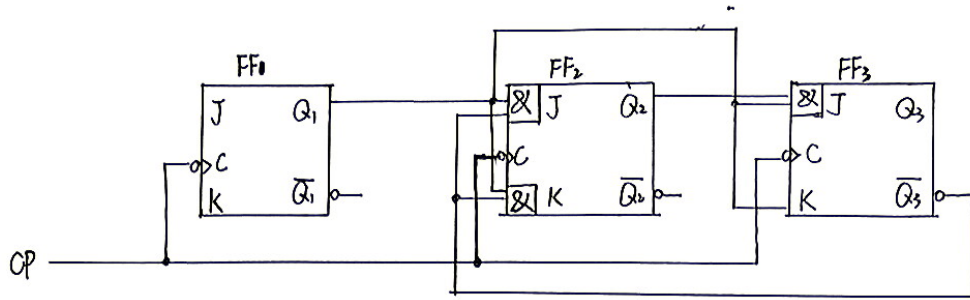
$$g = \overline{G_3} G_1 + G_2 G_0 + G_3 \overline{G_1} + G_3 G_2 + \overline{G_2} \overline{G_1} \overline{G_0} + \overline{G_2} G_1 G_0$$

电路图

第三章

分析时序电路的逻辑功能

6.



解:

这是同步时序逻辑电路, 时钟方程: $CP = CP_1 = CP_2 = CP_3$

没有输入和输出, 是 Moore 型电路

驱动方程:

$$\begin{cases} J_1 = 1 & K_1 = 1 \\ J_2 = Q_1^n \bar{Q}_3^n & K_2 = Q_1^n Q_3^n \\ J_3 = Q_1^n Q_2^n & K_3 = Q_1^n \end{cases}$$

将驱动方程代入 JK 触发器的特性方程 $Q^{n+1} = J\bar{Q}^n + \bar{K}Q^n$, 得

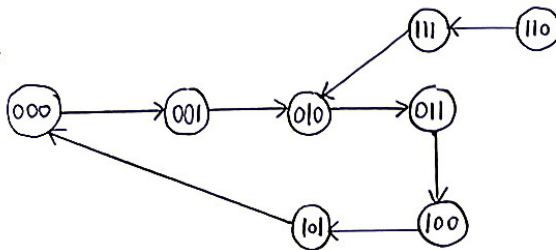
状态方程:

$$\begin{cases} Q_1^{n+1} = J_1\bar{Q}_1^n + \bar{K}_1Q_1^n = \bar{Q}_1^n \\ Q_2^{n+1} = J_2\bar{Q}_2^n + \bar{K}_2Q_2^n = Q_1^n\bar{Q}_2^n\bar{Q}_3^n + \bar{Q}_1^n\bar{Q}_3^nQ_2^n \\ Q_3^{n+1} = J_3\bar{Q}_3^n + \bar{K}_3Q_3^n = Q_1^nQ_2^n\bar{Q}_3^n + \bar{Q}_1^nQ_3^n \end{cases}$$

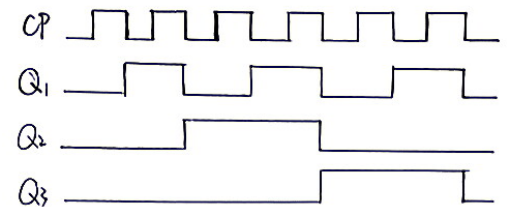
状态表:

现态			次态		
Q_3^n	Q_2^n	Q_1^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	0

状态转移图:



时序图:

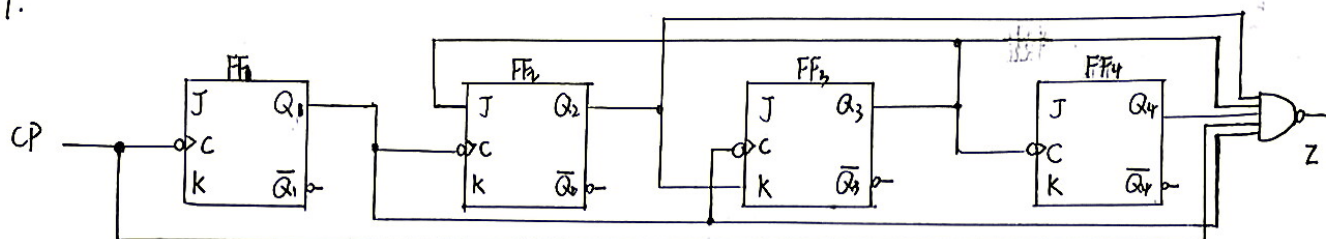


功能: 在输入第6个时钟脉冲 CP 后, 返回初始状态, 这是一个同步六进制计数器

自启动检查: 电路应有 $2^3 = 8$ 个工作状态, 只有6个有效状态, 还有2个无效状态 110 和 111,

如果电路由于某种原因进入无效状态, 只需继续输入 CP, 就能自动返回有效状态, 可以自启动

7.



解:

这是异步时序逻辑电路, 时钟方程: $CP_1 = CP, CP_2 = Q_1, CP_3 = Q_2, CP_4 = Q_3$

只有输出没有输入, 是 Moore 型电路

$$\text{驱动方程: } \begin{cases} J_1 = 1, K_1 = 1 \\ J_2 = Q_3, K_2 = 1 \\ J_3 = 1, K_3 = Q_2 \\ J_4 = 1, K_4 = 1 \end{cases}$$

将驱动方程代入 JK 触发器的特性方程, 得状态方程:

$$\begin{cases} Q_1^{n+1} = \overline{Q_1^n} & CP \downarrow \\ Q_2^{n+1} = Q_3^n \overline{Q_2^n} & Q_1^n \downarrow \\ Q_3^{n+1} = \overline{Q_3^n} + \overline{Q_2^n} Q_3^n & Q_2^n \downarrow \\ Q_4^{n+1} = \overline{Q_4^n} & Q_3^n \downarrow \end{cases}$$

$$\text{输出方程: } Z = \overline{Q_1 Q_2 Q_3 Q_4} CP$$

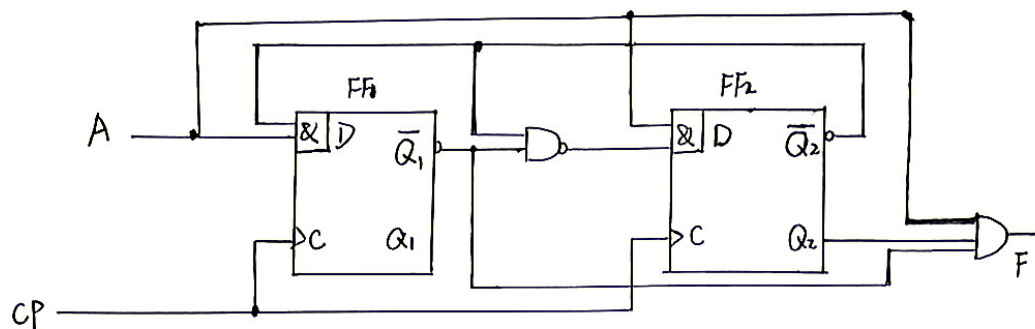
状态转移表:

现态 $Q_4^n Q_3^n Q_2^n Q_1^n$	次态 $Q_4^{n+1} Q_3^{n+1} Q_2^{n+1} Q_1^{n+1}$	输出 Z
0 0 0 0	0 0 1 1	0
0 0 0 1	0 0 0 0	0
0 0 1 0	0 0 1 1	0
0 0 1 1	0 0 0 0	0
0 1 0 0	0 0 1 1	0
0 1 0 1	0 1 0 0	0
0 1 1 0	0 0 1 1	0
0 1 1 1	1 0 0 0	0
1 0 0 0	1 0 1 1	0
1 0 0 1	1 0 0 0	0
1 0 1 0	1 0 1 1	0
1 0 1 1	1 0 0 0	0
1 1 0 0	1 0 1 1	0
1 1 0 1	1 1 0 0	0
1 1 1 0	1 0 1 1	0
1 1 1 1	1 0 0 0	1

状态转移图:

可以自启动

8.



解:

这是一个同步时序逻辑电路, 时钟方程: $CP = CP_1 = CP_2$
输出与输入和现态有关, 是 Mealy 型电路

$$\text{驱动方程: } \begin{cases} D_1 = A \cdot \overline{Q_2^n} \\ D_2 = \overline{Q_1^n} \cdot \overline{Q_2^n} \cdot A = (Q_1^n + Q_2^n) A \end{cases}$$

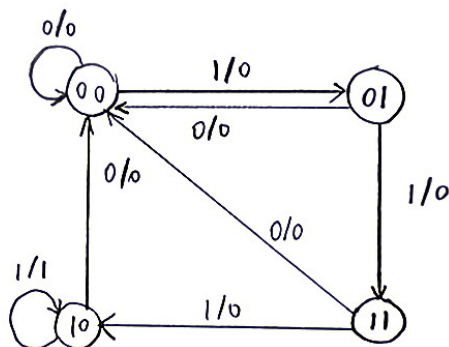
将驱动方程代入 D 触发器的特性方程 $Q^{n+1} = D$, 得状态方程:
$$\begin{cases} Q_1^{n+1} = D_1 = A \cdot \overline{Q_2^n} \\ Q_2^{n+1} = D_2 = A \cdot Q_1^n + A \cdot Q_2^n \end{cases}$$

$$\text{输出方程: } F = A \cdot Q_2^n \cdot \overline{Q_1^n}$$

状态转移表:

A	$Q_2^n Q_1^n$	$Q_2^{n+1} Q_1^{n+1}$	F
0	0 0	0 0	0
0	0 1	0 0	0
0	1 0	0 0	0
0	1 1	0 0	0
1	0 0	0 1	0
1	0 1	1 1	0
1	1 0	1 0	1
1	1 1	1 0	0

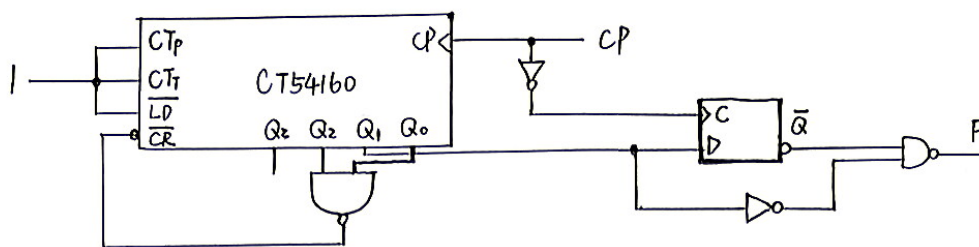
状态转移图:



功能: 检测序列 111

没有无效状态, 可以自启动

9.



解:

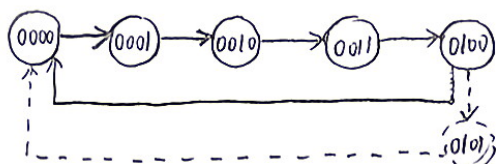
由54160功能表可知 CR为异步清零端

由图知 $CT_p = CT_i = 1$ ，处于计数状态。 $\overline{LD} = 1$ ，同步置数端高电平无效

$$\overline{CR} = \overline{Q_3^n Q_2^n Q_1^n Q_0^n}$$

表明，当 $Q_3^n = 1, Q_0^n = 1$ 且计数器输出0101时， $\overline{CR}$ 有效，使计数器清零 $\therefore \overline{CR}$ 为异步清零端，所以状态0101为暂态。

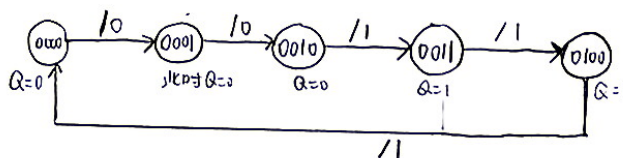
计数器的状态转移图为：

驱动方程: $D = Q_1^n$ 代入D触发器的特性方程，得 $Q^{n+1} = D = Q_1^n$ 输出方程: $F = \overline{Q_3^n \cdot Q_0^n} = Q_3^n + Q_0^n$

状态转移表:

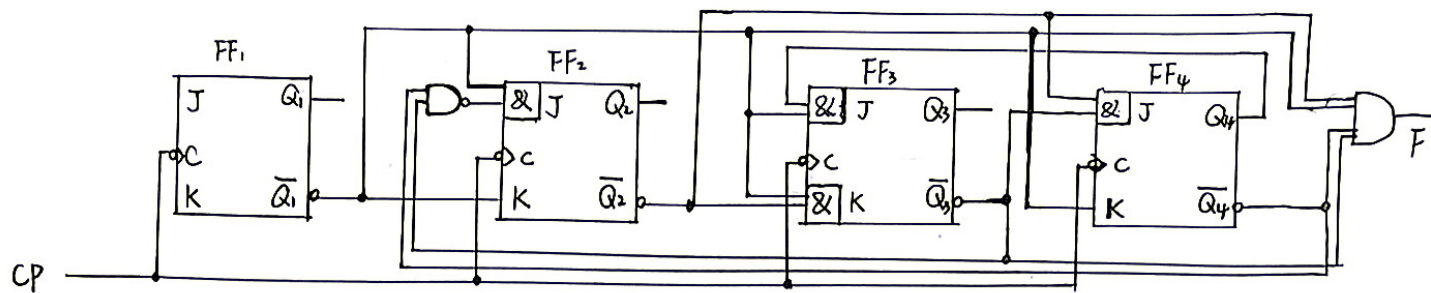
Q_3^n	Q_2^n	Q_1^n	Q_0^n	Q^{n+1}	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	Q^{n+1}	F
0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	0	0	0
0	0	1	0	0	0	0	0	1	0	0
0	0	1	1	0	0	0	1	1	0	1
0	1	0	0	1	0	1	0	0	1	1
0	1	0	1	1	0	1	0	1	1	1
0	1	1	0	1	0	1	1	0	1	1
0	1	1	1	1	0	1	1	1	1	1

状态转移图:



功能?

自启动?



解: 这是一个同步时序逻辑电路, 时钟方程: $CP = CP_1 = CP_2 = CP_3 = CP_4$
输出仅与现态有关, 是 Moore 型电路

驱动方程

$$\begin{cases} J_1 = 1 & K_1 = 1 \\ J_2 = \overline{Q_1^n} \cdot \overline{Q_3^n} \cdot \overline{Q_4^n} & K_2 = \overline{Q_1^n} \\ J_3 = Q_4^n \cdot \overline{Q_1^n} & K_3 = \overline{Q_2^n} \cdot \overline{Q_1^n} \\ J_4 = \overline{Q_2^n} \cdot \overline{Q_3^n} & K_4 = \overline{Q_1^n} \end{cases}$$

将驱动方程代入 JK 触发器的特征方程, 得状态转移方程

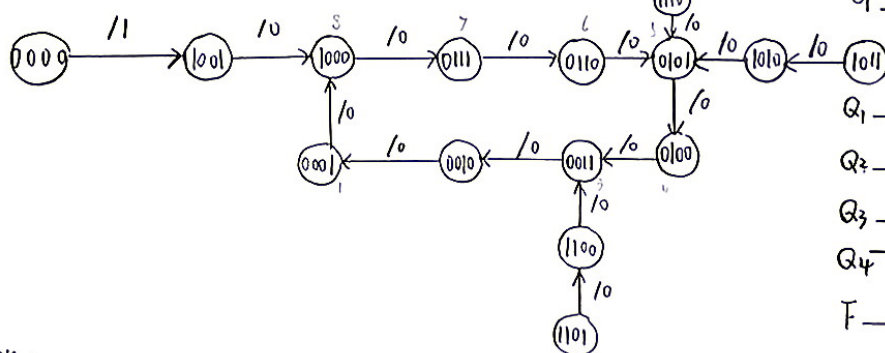
$$\begin{cases} Q_1^{n+1} = J_1 \overline{Q_1^n} + \overline{K_1} Q_1^n = \overline{Q_1^n} \\ Q_2^{n+1} = J_2 \overline{Q_2^n} + \overline{K_2} Q_2^n = \overline{Q_1^n} \cdot (\overline{Q_3^n} + \overline{Q_4^n}) \overline{Q_2^n} + Q_1^n Q_2^n \\ Q_3^{n+1} = J_3 \overline{Q_3^n} + \overline{K_3} Q_3^n = Q_4^n \overline{Q_1^n} \overline{Q_3^n} + (\overline{Q_2^n} + Q_1^n) Q_3^n \\ Q_4^{n+1} = J_4 \overline{Q_4^n} + \overline{K_4} Q_4^n = \overline{Q_2^n} \overline{Q_3^n} \overline{Q_4^n} + Q_1^n Q_4^n \end{cases}$$

输出方程: $F = \overline{Q_1^n} \overline{Q_2^n} \overline{Q_3^n} \overline{Q_4^n} = \overline{Q_1^n + Q_2^n + Q_3^n + Q_4^n}$

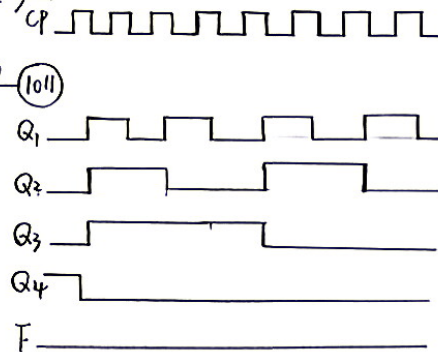
状态表:

Q_4^n	Q_3^n	Q_2^n	Q_1^n	Q_4^{n+1}	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	F
✓	0	0	0	1	0	0	1	1
✓	0	0	1	1	0	0	0	0
✓	0	1	0	0	0	0	1	0
✓	0	1	1	0	0	1	0	0
✓	0	1	0	0	0	1	1	0
✓	0	1	1	0	1	0	0	0
✓	0	1	1	0	1	0	1	0
✓	1	0	0	0	1	1	0	0
✓	1	0	0	1	0	0	0	0
✓	1	0	1	0	0	0	1	0
✓	1	0	1	1	0	1	0	0
✓	1	1	0	0	0	1	0	0
✓	1	1	0	1	1	0	0	0
✓	1	1	1	0	1	1	1	0
✓	1	1	1	1	1	1	1	0

状态转移图:



波形图:



功能:

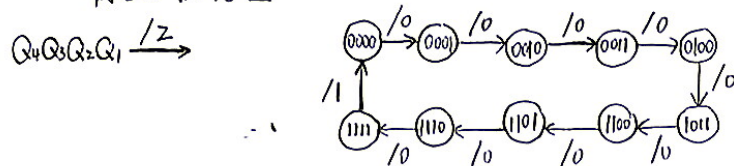
可以自启动

设计:

13. 设计2421码十进制同步计数器

解:

原始状态图



已是最简, 状态编码已完成

状态转移真值表

$Q_4 Q_3 Q_2 Q_1$	$Q_4^{n+1} Q_3^{n+1} Q_2^{n+1} Q_1^{n+1}$	Z
0000	0001	0
0001	0010	0
0010	0011	0
0011	0100	0
0100	0101	0
0101	0110	0
0110	0111	0
0111	1000	0
1000	1001	0
1001	1010	1
1010	1011	0
1011	1100	0
1100	1101	0
1101	1110	0
1110	1111	0
1111	0000	0

选用4个JK触发器 FF_1, FF_2, FF_3, FF_4

由JK触发器激励表

$Q^n \rightarrow Q^{n+1}$	J	K
0 → 0	0	d
0 → 1	1	d
1 → 0	d	1
1 → 1	d	0

得激励函数真值表

J_1	J_2	J_3	J_4
0	0	0	0
0	1	0	0
0	0	0	0
0	0	0	0
1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0

$Q_4 Q_3$	$Q_2 Q_1$	J_1
00	00	0
00	01	0
00	10	0
00	11	0
01	00	0
01	01	0
01	10	0
01	11	0
10	00	1
10	01	1
10	10	1
10	11	1

$Q_4 Q_3$	$Q_2 Q_1$	K_1
00	00	d
00	01	d
00	10	d
00	11	d
01	00	d
01	01	d
01	10	d
01	11	d
10	00	d
10	01	d
10	10	d
10	11	d

$Q_4 Q_3$	$Q_2 Q_1$	J_2
00	00	0
00	01	0
00	10	0
00	11	0
01	00	0
01	01	0
01	10	0
01	11	0
10	00	1
10	01	1
10	10	1
10	11	1

$Q_4 Q_3$	$Q_2 Q_1$	K_2
00	00	d
00	01	d
00	10	d
00	11	d
01	00	d
01	01	d
01	10	d
01	11	d
10	00	1
10	01	1
10	10	1
10	11	1

$Q_4 Q_3$	$Q_2 Q_1$	J_3
00	00	0
00	01	0
00	10	0
00	11	0
01	00	0
01	01	0
01	10	0
01	11	0
10	00	1
10	01	1
10	10	1
10	11	1

$Q_4 Q_3$	$Q_2 Q_1$	K_3
00	00	d
00	01	d
00	10	d
00	11	d
01	00	d
01	01	d
01	10	d
01	11	d
10	00	d
10	01	d
10	10	d
10	11	d

$Q_4 Q_3$	$Q_2 Q_1$	J_4
00	00	0
00	01	0
00	10	0
00	11	0
01	00	0
01	01	0
01	10	0
01	11	0
10	00	1
10	01	1
10	10	1
10	11	1

$Q_4 Q_3$	$Q_2 Q_1$	K_4
00	00	d
00	01	d
00	10	d
00	11	d
01	00	d
01	01	d
01	10	d
01	11	d
10	00	1
10	01	1
10	10	1
10	11	1

得驱动方程:

$$\begin{cases} J_1 = 1 & K_1 = 1 \\ J_2 = Q_1^n & K_2 = Q_1^n \\ J_3 = Q_2^n Q_1^n & K_3 = \overline{Q_4^n} + Q_2^n Q_1^n \\ J_4 = Q_3^n & K_4 = Q_3^n Q_2^n Q_1^n \end{cases}$$

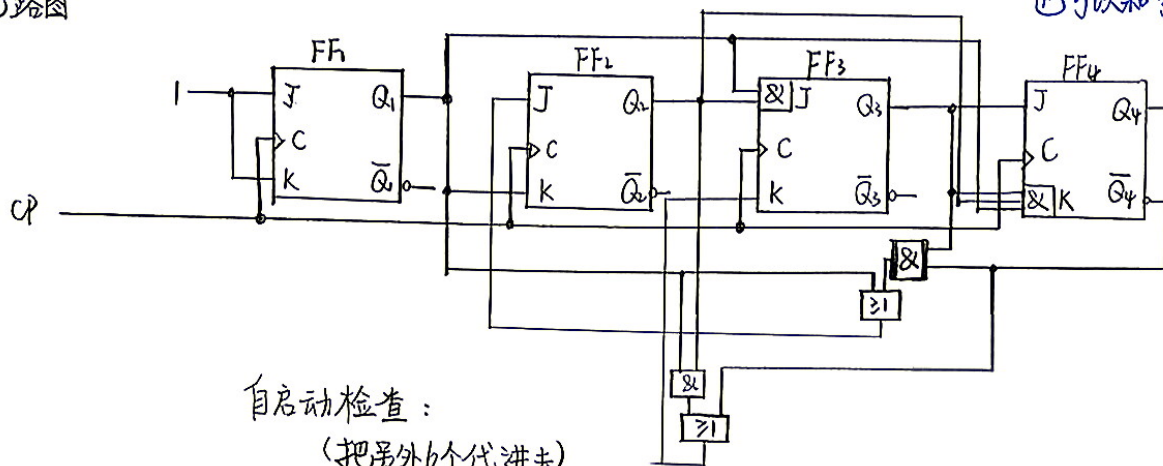
输出方程: $Z = Q_4^n Q_3^n Q_2^n Q_1^n$

画卡诺图!

就算只有一个状态,

也可以和可能存在的无效状态合并化简

逻辑电路图



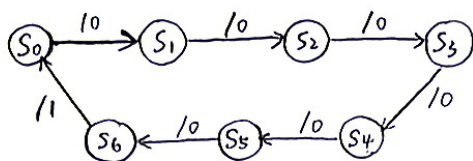
自启动检查:

(把另外6个代进去)

14. 设计模7同步计数器

解:

原始状态图:



已最简

对状态进行编码: $S_0=000, S_1=001, S_2=010, S_3=011, S_4=100, S_5=101, S_6=110$

假设选用3个D触发器来说明

状态转移真值表

$Q_3^n Q_2^n Q_1^n$	$Q_3^{n+1} Q_2^{n+1} Q_1^{n+1}$	Z
0 0 0	0 0 1	0
0 0 1	0 1 0	0
0 1 0	0 1 1	0
0 1 1	1 0 0	0
1 0 0	1 0 1	0
1 0 1	1 1 0	0
1 1 0	0 0 0	1

用D触发器激励表

$Q^n \rightarrow Q^{n+1}$	D
0 0	0
0 1	1
1 0	0
1 1	1

得激励函数真值表

D_3	D_2	D_1
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
0	0	0

$Q_3^n \backslash Q_2^n Q_1^n$	00	01	11	10
0	0	0	1	0
1	1	1	d	0

D_3

$Q_3^n \backslash Q_2^n Q_1^n$	00	01	11	10
0	0	1	0	1
1	0	1	d	0

D_2

$Q_3^n \backslash Q_2^n Q_1^n$	00	01	11	10
0	1	0	0	1
1	1	0	d	0

D_1

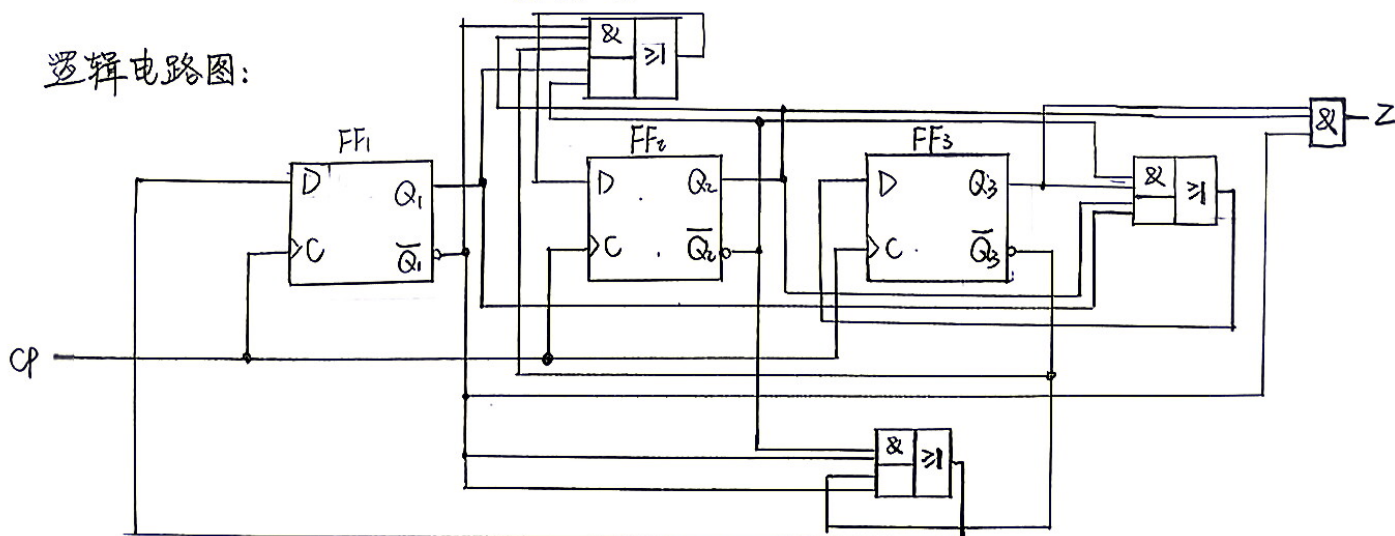
驱动方程

$$\begin{cases} D_3 = Q_3^n \bar{Q}_2^n + Q_2^n Q_1^n \\ D_2 = \bar{Q}_3^n \bar{Q}_2^n \bar{Q}_1^n + \bar{Q}_3^n Q_1^n \\ D_1 = \bar{Q}_3^n \bar{Q}_1^n + \bar{Q}_2^n \bar{Q}_1^n = (\bar{Q}_3^n + \bar{Q}_2^n) \bar{Q}_1^n \\ \quad = \overline{Q_3^n \cdot Q_2^n} \cdot \bar{Q}_1^n \end{cases}$$

状态方程: $\begin{cases} Q_3^{n+1} = Q_3^n \bar{Q}_2^n + Q_2^n Q_1^n \\ Q_2^{n+1} = \bar{Q}_3^n \bar{Q}_2^n \bar{Q}_1^n + \bar{Q}_3^n Q_1^n \\ Q_1^{n+1} = \bar{Q}_3^n \bar{Q}_1^n + \bar{Q}_2^n \bar{Q}_1^n \end{cases}$

输出方程: $Z = Q_3^n Q_2^n \bar{Q}_1^n$

逻辑电路图:



自启动检查: 将无效状态 111 代入状态方程, 得其次态为 100,

即: 当电路因某种原因进入无效状态, 在下一个时钟脉冲到来后, 可自动进入有效状态, 可以自启动

15. 按下列给定状态转移表, 设计同步计数器

序号	A	B	C	D
0	0	0	0	0
1	0	0	0	1
2	0	1	0	0
3	0	1	0	1
4	0	1	1	0
5	0	1	1	1
6	1	0	0	0
7	1	0	0	1
8	1	1	0	0
9	1	1	0	1
10	1	1	1	0
11	1	1	1	1

解: 设计一个自启动电路:

由JK触发器的激励表 可得下表:

$Q^n \rightarrow Q^{n+1}$	J	K
0 0	0	d
0 1	1	d
1 0	d	1
1 1	d	0

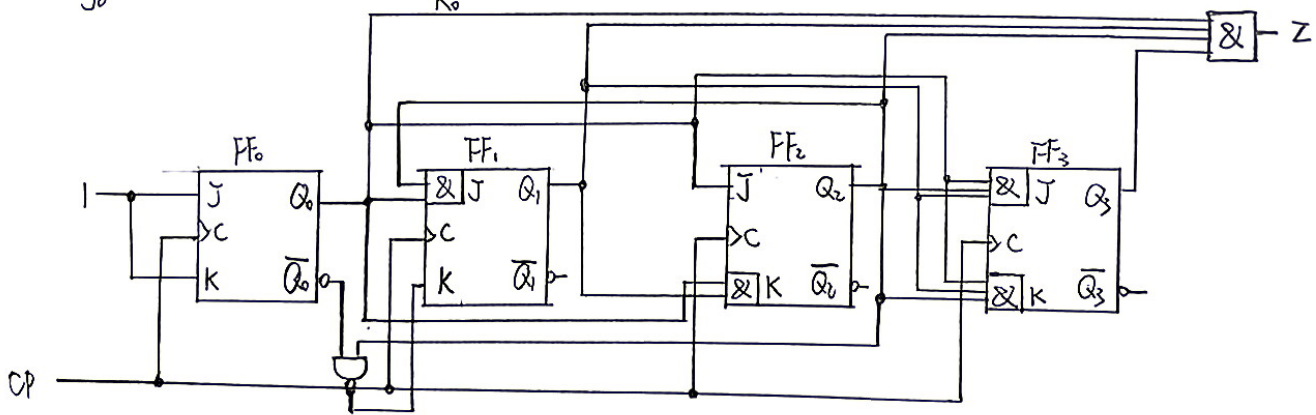
$Q_3^n Q_2^n Q_1^n Q_0^n$	$Q_3^{n+1} Q_2^{n+1} Q_1^{n+1} Q_0^{n+1}$	$J_3 K_3$	$J_2 K_2$	$J_1 K_1$	$J_0 K_0$	Z
0 0 0 0	0 0 0 1	0 d	0 d	0 d	1 d	0
0 0 0 1	0 1 0 0	0 d	1 d	0 d	d 1	0
0 0 1 0	0 1 0 1	0 d	d 0	0 d	1 d	0
0 0 1 1	0 1 1 0	0 d	d 0	1 d	d 1	0
0 1 0 0	1 0 0 0	1 d	d 1	d 1	d 1	0
0 1 0 1	1 0 0 1	d 0	0 d	0 d	1 d	0
0 1 1 0	1 1 0 0	d 0	1 d	0 d	d 1	0
0 1 1 1	1 1 0 1	d 0	d 0	0 d	1 d	0
1 0 0 0	1 1 1 1	d 1	d 1	d 1	d 1	0
1 0 0 1	1 1 1 0	d 1	d 1	d 1	d 1	0
1 0 1 0	0 0 0 1	0 d	0 d	d 1	1 d	0
1 0 1 1	0 1 0 0	0 d	1 d	d 1	d 1	0
1 1 0 0	1 0 0 1	1 d	0 d	d 1	1 d	0
1 1 0 1	1 1 0 0	1 d	1 d	d 1	d 1	0
1 1 1 0	1 1 1 1	1 d	1 d	d 1	d 1	0
1 1 1 1	0 0 0 0	d 1	d 1	d 1	d 1	1

输出方程: $Z = Q_3^n Q_2^n Q_1^n Q_0^n$

驱动方程:

$$\begin{cases} J_3 = Q_2^n Q_1^n Q_0^n & K_3 = Q_2^n Q_1^n Q_0^n \\ J_2 = Q_0^n & K_2 = Q_1^n Q_0^n \\ J_1 = Q_2^n Q_0^n & K_1 = \overline{Q_2^n} + Q_0^n = \overline{Q_2^n} \cdot \overline{Q_0^n} \\ J_0 = 1 & K_0 = 1 \end{cases}$$

$Q_3^n Q_2^n$	Q_1^n	Q_0^n	J_3	K_3	J_2	K_2	J_1	K_1	J_0	K_0
00	0	0	0	1	0	1	0	1	1	1
00	0	1	0	1	0	1	0	1	1	1
00	1	0	0	1	0	1	0	1	1	1
00	1	1	0	1	0	1	0	1	1	1
01	0	0	0	1	0	1	0	1	1	1
01	0	1	0	1	0	1	0	1	1	1
01	1	0	0	1	0	1	0	1	1	1
01	1	1	0	1	0	1	0	1	1	1
10	0	0	0	1	0	1	0	1	1	1
10	0	1	0	1	0	1	0	1	1	1
10	1	0	0	1	0	1	0	1	1	1
10	1	1	0	1	0	1	0	1	1	1



20. 设计一个可控同步计数器, M_1, M_2 为控制信号

解. 状态转移图:

状态转移表:

$Q_3^n Q_2^n Q_1^n$	$Q_3^{n+1} Q_2^{n+1} Q_1^{n+1}$	Z
0 0 0	0 0 1	0
0 0 1	0 1 0	1
0 1 0	0 1 1	0
0 1 1	1 0 0	1
1 0 0	1 0 1	0
1 0 1	1 1 0	0
1 1 0	1 1 1	0
1 1 1	0 0 0	1

次态卡诺图:

$Q_3^n \backslash Q_2^n Q_1^n$	00	01	11	10
$Q_3^n = 0$	0	0	1	0
$Q_3^n = 1$	1	1	0	1

状态方程:

$$Q_3^{n+1} = Q_3^n \bar{Q}_2^n + Q_3^n \bar{Q}_1^n + Q_3^n Q_2^n Q_1^n$$

$Q_3^n \backslash Q_2^n Q_1^n$	00	01	11	10
$Q_3^n = 0$	0	1	0	1
$Q_3^n = 1$	0	1	0	1

$$Q_2^{n+1} = \bar{Q}_2^n Q_1^n + Q_2^n \bar{Q}_1^n$$

$Q_3^n \backslash Q_2^n Q_1^n$	00	01	11	10
$Q_3^n = 0$	1	0	0	1
$Q_3^n = 1$	1	0	0	1

$$Q_1^{n+1} = \bar{Q}_1^n$$

与JK触发器的特性方程 $Q^{n+1} = J\bar{Q}^n + KQ^n$, 得驱动方程

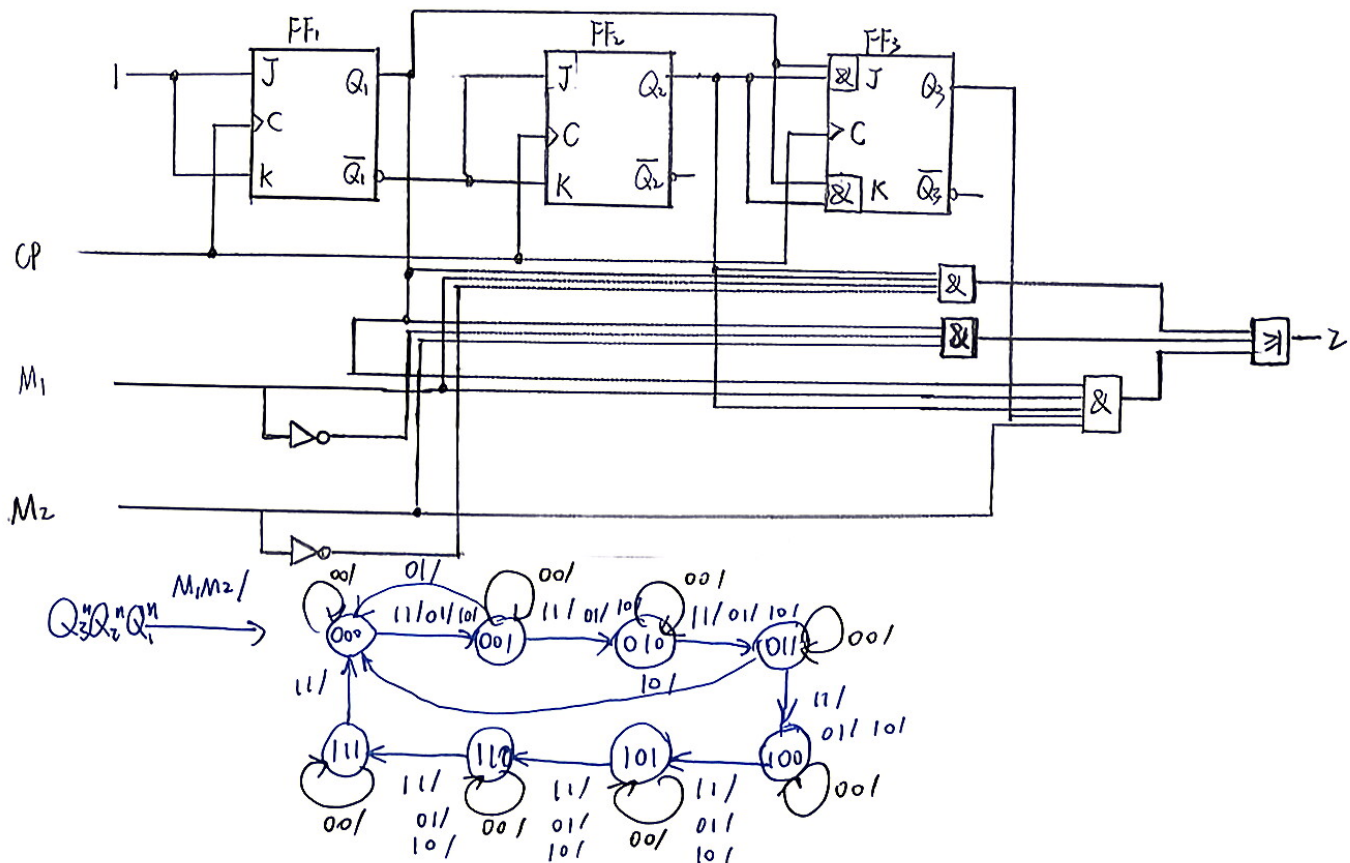
$$\begin{cases} J_3 = Q_1^n Q_2^n & K_3 = Q_1^n Q_2^n \\ J_2 = \bar{Q}_1^n & K_2 = \bar{Q}_1^n \\ J_1 = 1 & K_1 = 1 \end{cases}$$

实现

$M_1 M_2 = 01, 10, 11$ 分别在 $Q_3^n Q_2^n Q_1^n = 001, 011, 111$ 输出, 分别实现模2, 4, 8计数

$Q_3^n \backslash Q_2^n Q_1^n$	00	01	11	10
$Q_3^n = 0$	0	1	1	0
$Q_3^n = 1$	0	0	1	0

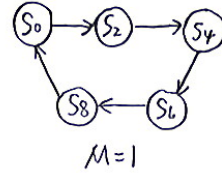
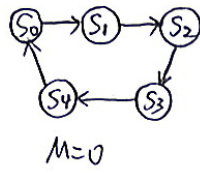
$$Z = \bar{Q}_3^n Q_1^n + Q_2^n Q_1^n$$



21. 设计一个用M信号控制的五进制同步计数器, 要求: $M=0$ 且有CP, 加1顺序计数; $M=1$ 且有CP, 加2顺序计数

解: 根据题目要求, 由于最大计数模值是9, 采用4个触发器

原始状态图



令 $S_0=0000$, $S_1=0001$, $S_2=0010$, $S_3=0011$, $S_4=0100$, $S_5=0101$, $S_6=0110$, $S_7=0111$, $S_8=1000$

状态转移表

$M=0$					$M=1$				
Q_4^n	Q_3^n	Q_2^n	Q_1^n		Q_4^{n+1}	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	
0	0	0	0	0	0	0	0	1	
0	0	0	0	1	0	0	1	0	
0	0	0	1	0	0	0	1	1	
0	0	0	1	1	0	1	0	0	
0	1	0	0	0	0	0	0	0	
0	1	0	0	1	X	X	X	X	
0	1	1	0	0					
0	1	1	0	1					
1	0	0	0	0					
1	0	0	0	1					
1	0	1	0	0					
1	0	1	0	1					
1	1	0	0	0					
1	1	0	0	1					
1	1	1	0	0					
1	1	1	0	1					
1	1	1	1	0					
1	1	1	1	1					

次态卡诺图:

$M=0$ 时:

$M=1$ 时:

当 $M=0$ 时:

$$\begin{cases} Q_4^{n+1}=0 \\ Q_3^{n+1}=Q_2^n Q_1^n \\ Q_2^{n+1}=\overline{Q_2^n} Q_1^n + Q_2^n \overline{Q_1^n} \\ Q_1^{n+1}=\overline{Q_3^n} Q_1^n \end{cases}$$

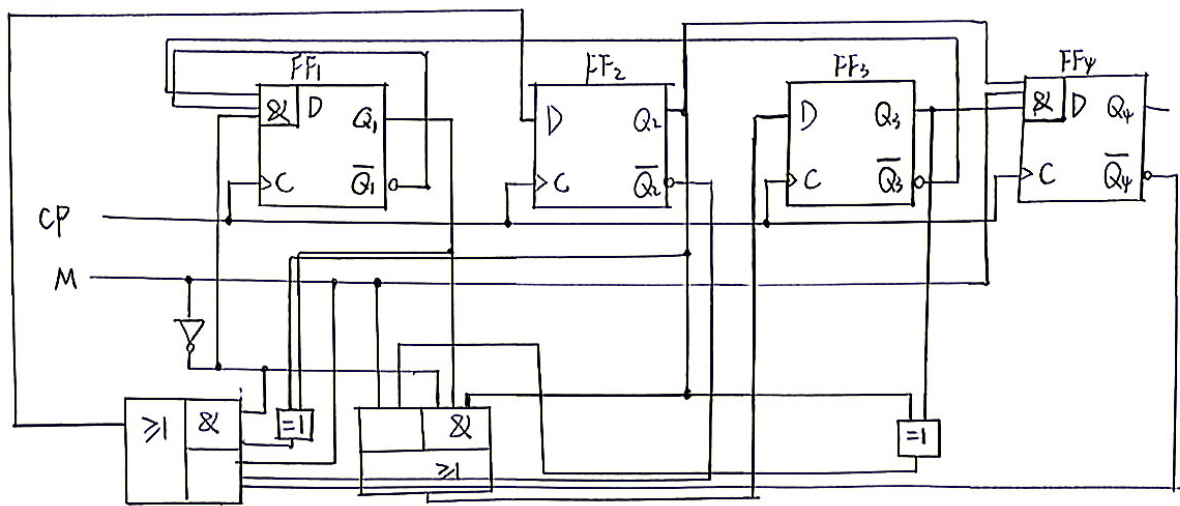
当 $M=1$ 时:

$$\begin{cases} Q_4^{n+1}=Q_3^n Q_2^n \\ Q_3^{n+1}=\overline{Q_2^n} Q_2^n + Q_3^n \overline{Q_2^n} \\ Q_2^{n+1}=\overline{Q_4^n} \overline{Q_3^n} \\ Q_1^{n+1}=0 \end{cases}$$

综合考虑:

$$\begin{cases} Q_4^{n+1}=\overline{M} Q_3^n Q_2^n = D_4 \\ Q_3^{n+1}=\overline{M} Q_2^n Q_1^n + M(\overline{Q_2^n} Q_2^n + Q_3^n \overline{Q_2^n}) = D_3 \\ Q_2^{n+1}=\overline{M}(\overline{Q_4^n} \overline{Q_3^n} + Q_2^n \overline{Q_1^n}) + M \overline{Q_4^n} \overline{Q_3^n} = D_2 \\ Q_1^{n+1}=\overline{M} \overline{Q_3^n} Q_1^n = D_1 \end{cases}$$

逻辑电路图:



用D触发器

22. 对下列原始状态转移表进行化简, 并设计电路

(1)

$S(t)$	$N(t)$		$Z(t)$	
	$X=0$	$X=1$	$X=0$	$X=1$
A	A	B	0	0
B	C	A	0	1
C	B	D	0	1
D	D	C	0	0

解:

B	X		
C	X	BC AD	
D	AD BC	X	X
	A	B	C

A' B'
 $(A,D), (B,C)$ 为等价类且已是最大等价类

化简后的状态表

$S(t)$	$N(t)$		$Z(t)$	
	$X=0$	$X=1$	$X=0$	$X=1$
A'	A'	B'	0	0
B'	B'	A'	0	1

状态编码: 只有2个状态, 用位即可, 令 $A'=0, B'=1$

状态转移表

X	Q^n	Q^{n+1}	Z^n
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

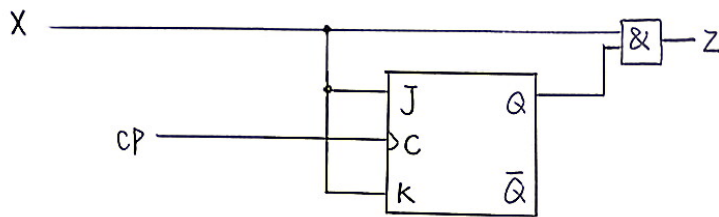
X	Q^n	0	1
0		0	1
1		1	0
	Q^{n+1}		

状态方程: $Q^{n+1} = \bar{X}Q^n + XQ^n$

设用JK触发器, 则驱动方程: $J=X, K=X$

输出方程: $Z=XQ^n$

X	Q^n	0	1
0		0	0
1		0	1
	Z		



(2)

S(t)	N(t)		Z(t)	
	X=0	X=1	X=0	X=1
A	B	H	0	0
B	E	C	0	1
C	D	F	0	0
D	G	A	0	1
E	A	H	0	0
F	E	B	1	1
G	C	F	0	0
H	G	D	1	1

解:

B	X						
C	BD HF	X					
D	X	EG CA	X				
E	AB	X	DA EH	X			
F	X	X	X	X	X		
G	BC HE	X	DC	X	AG HF	X	
H	X	X	X	X	X	EG BD	X
	A	B	C	D	E	F	G

可找到等价类(B,D), (H,F), (A,C), (E,G), 已是最大等价类
最简状态表:

S(t)	N(t)		Z(t)	
	X=0	X=1	X=0	X=1
S ₁	S ₂	S ₄	0	0
S ₂	S ₃	S ₁	0	1
S ₃	S ₁	S ₄	0	0
S ₄	S ₃	S ₂	1	1

状态编码: 4个状态, 用2个编码即可.

令 $S_1 = Q_2^n Q_1^n = 00$, $S_2 = Q_2^n Q_1^n = 01$, $S_3 = Q_2^n Q_1^n = 10$, $S_4 = Q_2^n Q_1^n = 11$

状态转移表:

X	$Q_2^n Q_1^n$		$Q_2^{n+1} Q_1^{n+1}$		Z^n
0	0	0	0	1	0
0	0	1	1	0	0
0	1	0	0	0	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	0	0	1
1	1	0	1	1	0
1	1	1	0	1	1

$Q_2^n Q_1^n$	00	01	11	10
X=0	0	0	1	0
X=1	0	1	1	0

Z

设选用JK触发器:

$Q_2^n Q_1^n$	00	01	11	10
X=0	0	1	1	0
X=1	1	0	0	1

Q_2^{n+1}

$Q_2^n Q_1^n$	00	01	11	10
X=0	1	0	0	0
X=1	1	0	1	1

Q_1^{n+1}

得状态方程: $Q_2^{n+1} = \overline{Q_2^n} Q_1^n \overline{X} + Q_2^n Q_1^n \overline{X} + X Q_2^n \overline{Q_1^n} + X Q_2^n Q_1^n = (X \oplus Q_2^n) \overline{Q_1^n} + Q_2^n Q_1^n$
 $Q_1^{n+1} = \overline{Q_2^n} \overline{Q_1^n} + X Q_2^n Q_1^n + X \overline{Q_2^n} Q_1^n = (X + \overline{Q_2^n}) \overline{Q_1^n} + Q_2^n X Q_1^n$

驱动方程: $J_2 = X \oplus Q_1^n$, $K_2 = \overline{X \oplus Q_1^n}$
 $J_1 = X + \overline{Q_2^n}$, $K_1 = \overline{X Q_2^n}$
输出方程: $Z = X Q_1^n + Q_2^n Q_1^n$

